CSE 6242 POLYMARKET TRADER ANALYSIS

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Motivation

Problem

Prediction markets generate large volumes of trade data, but trader behavior and profitability patterns are difficult to interpret.

Why it matters

Understanding trader strategies helps evaluate market efficiency and identify meaningful behavioral patterns.

Objective

Build a pipeline to transform trade data into trader-level insights, identify behavioral clusters, and create meaningful visualizations.

Approach

- Transform trade-level data into trader-level features
- Normalize features for comparability
- Apply clustering (K-Means, DBSCAN)
- Visualize clusters using interactive D3 plots

Intuition & Contribution

- Traders with similar behaviors form natural groups (e.g., high-frequency vs. high-volume)
- Clustering reveals hidden structure in market participation

What's New

- Focus on trader-level behavior instead of market-level trends
- Identify trader archetypes (casual, specialists, bots, "bonders")
- Enable interactive exploration with user-selected feature axes

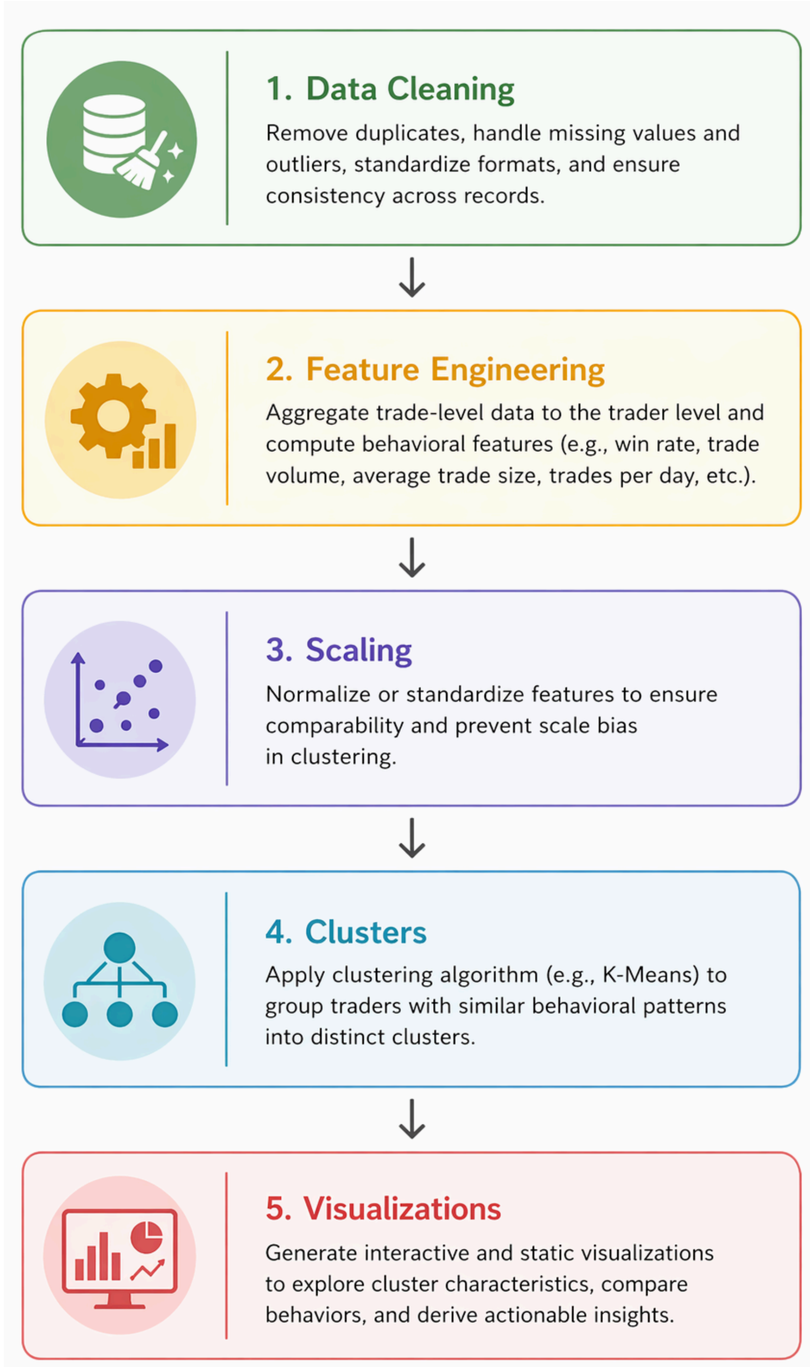

Figure 1: The pipeline for transforming data into clustered insights and visualizations of trader behavior

Data

- Polymarket trade and market data (downloaded dataset)
- Stored as Parquet files for efficient querying
- ~30GB dataset (large-scale, real-world data)
- >400 million trade records across >1.9 million traders and >400 thousand markets
- Temporal: includes timestamps for all trades
- Includes trader IDs, trade size, odds, outcomes

Data Processing

- Randomly sampled 1% of traders with >100 trades
- Queried using DuckDB for fast analytical processing
- Aggregated from trade-level → trader-level features
- Clustered using K-means and HDBSCAN, determined that K-means produces better clusters

Features

We transform raw trade data into trader-level behavioral features:

- Win rate
- Average trade size
- Total trade volume
- Number of trades
- Trades per day (frequency)
- Net gains/loss
- Average odds
- Preferred market category

These features capture both performance and behavior, enabling meaningful comparison across traders.

Results

- Using the features of win rate, average odds betted on, profit per trade, total number of trades, trade frequency, average trade size, and total trade volume allow for meaningful clustering
- Trader archetypes were primarily based on their behavior, existing wealth, and relative success
- Distinct groups of unsuccessful traders, safe traders with modest profit, and good traders were detected

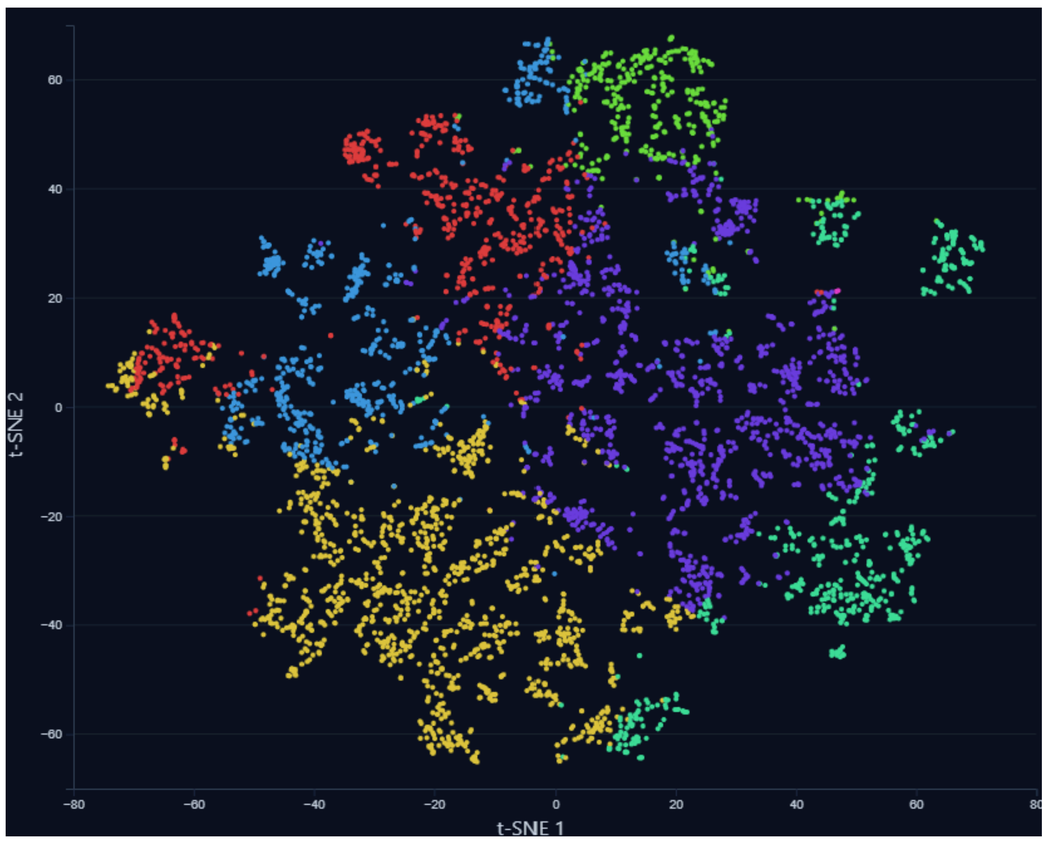

Figure 2: The default graph showing the t-SNE plot

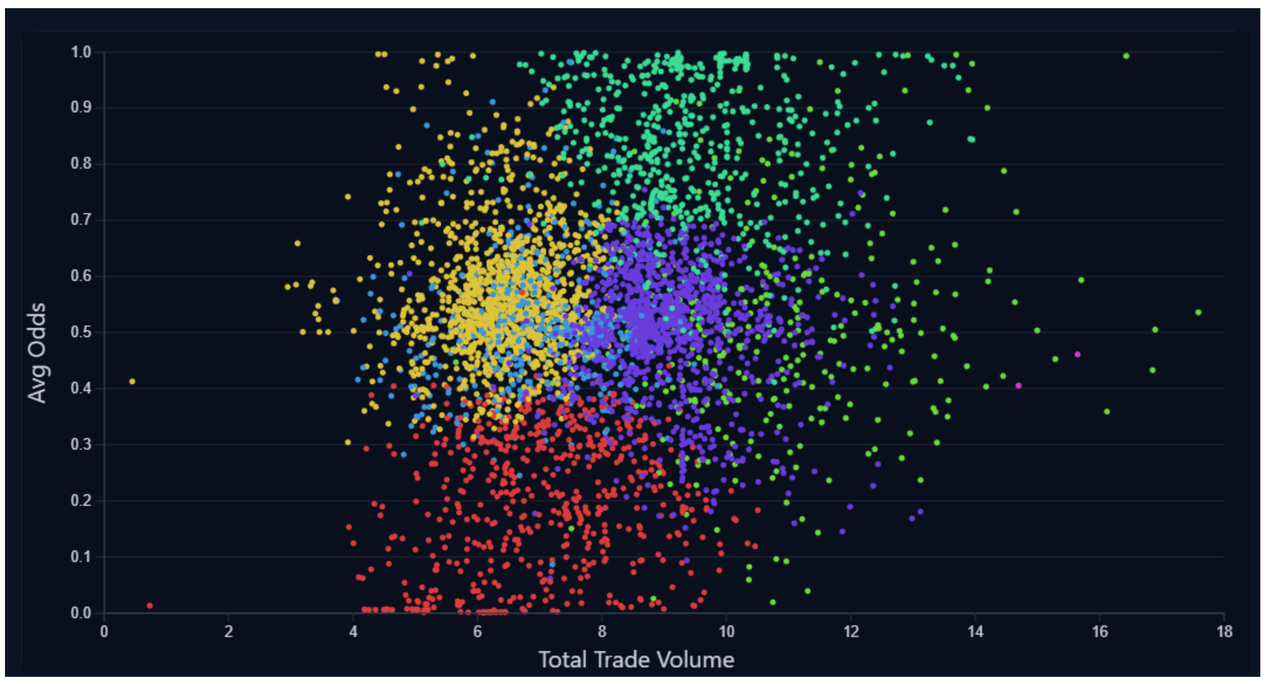

Figure 3: Display of traders for total trade volume vs. average odds

Conclusions and Future Directions

We built a pipeline that transforms raw trade data into interactive visualizations of trader behavior and archetypes. By combining feature engineering, clustering, and dynamic visual exploration, we reveal meaningful patterns in how traders operate and influence markets. This approach makes prediction markets more transparent and easier to understand. We can build on this pipeline by using larger data and smarter clustering appoaches and extend this to Kalshi Datasets as well for the future.